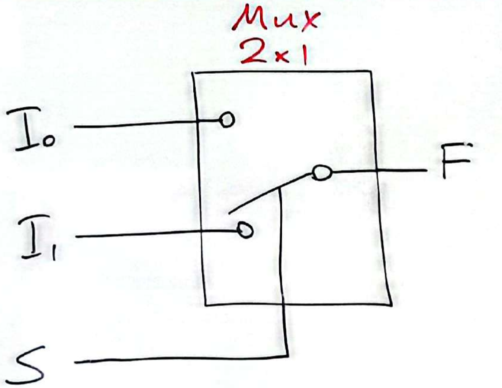
Mux (Multiplexers) – Veri Çoklayıcılar



$S=0 \Rightarrow F = I_0$

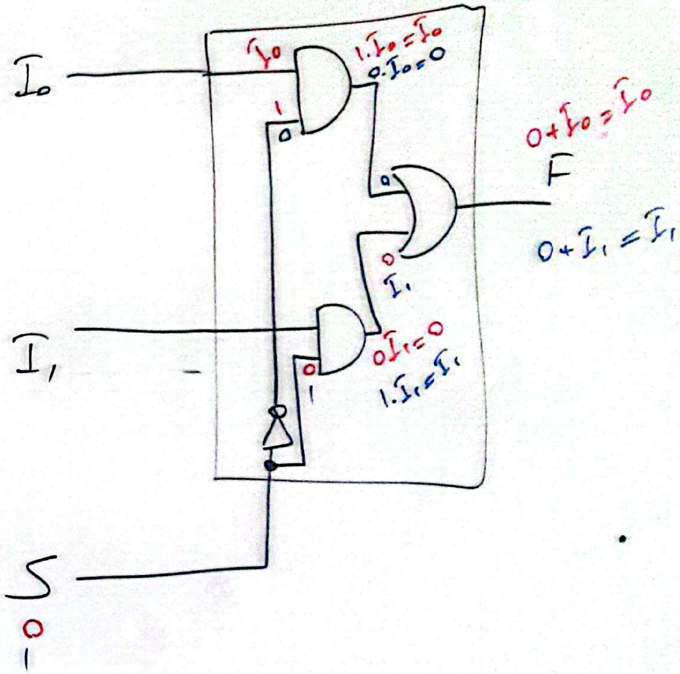
$S=1 \Rightarrow F = I_1$

S	I_1	I_0	F	
0	0	0	0	
0	0	1	1	$\rightarrow \bar{S}.\bar{I_1}.I_0$
0	1	0	0	
0	1	1	1	$\rightarrow \bar{S}.I_1.I_0$
1	0	0	0	
1	0	1	0	
1	1	0	1	$\rightarrow S.I_1.\bar{I_0}$
1	1	1	1	$\rightarrow S.I_1.I_0$

$\bar{S}.\bar{I_1}.I_0 + \bar{S}.I_1.I_0 = \bar{S}.I_0$

$S.I_1.\bar{I_0} + S.I_1.I_0 = S.I_1$

$F(S, I_1, I_0) = \bar{S}.I_0 + S.I_1$



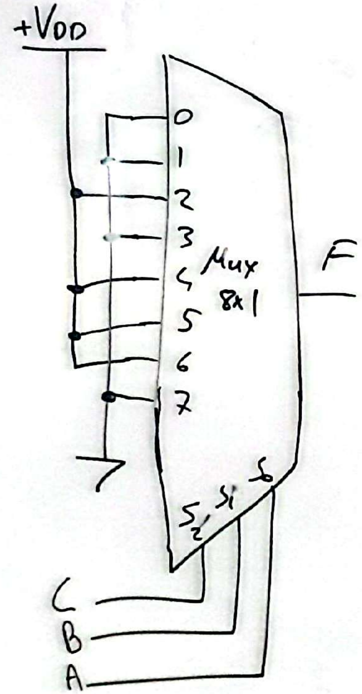
$F(C,B,A) = \Sigma(2,4,5,6)$

a) 8x1 Mux ile devreyi tasarlayınız.

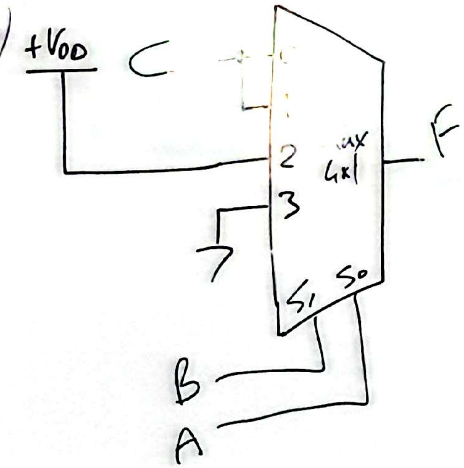
b) 4x1 " " " " $(S_1 = B,\ S_0 = A)$

c) 2x1 " " " " $(S_0 = C)$

a)



b)



	C	B	A	F	
0 →	0	0	0	0	C
1 →	0	0	1	0	C
2 →	0	1	0	1	1
3 →	0	1	1	0	0
0 →	1	0	0	1	
1 →	1	0	1	1	
2 →	1	1	0	1	
3 →	1	1	1	0	

c)

C	B	A	F	
0	0	0	0	
0	0	1	0	
0	1	0	1	→ $B.\bar{A}$
0	1	1	0	
1	0	0	1	
1	0	1	1	
1	1	0	1	
1	1	1	0	→ $\bar{B} + \bar{A}$